Zhao Tang
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EDUCATION:

JUN 2012-AUG 2020 (expected) — **Rice University, Houston, TX**
PhD in Chemical and Biomolecular Engineering, GPA 3.63/4.0 — Research Advisor: Prof. Matteo Pasquali
SEP 2009-JUN 2012 — **Purdue University, West Lafayette, IN**
BSE with honor degree in Chemical Engineering, GPA 3.76/4.00 — Research Advisor: Prof. Alex Wei

TECHNICAL SKILLS:

- Programming skills: Digital image/signal processing, Algorithm
- Numerical skills: Machine Learning, statistical data analysis, scientific programming, Numerical simulation, CFD
- Software skills: Matlab, C/C++, Fortran, Python, Comsol, AutoCAD, Visual Basic, ImageJ/Fiji, Illustrator, Photoshop
- Lab skills: Photolithography, microfluidics, Microscopy and single-molecular spectroscopy, Clean room certified, Rheology and fluid processing, micro-rheology;
- Laboratory instruments and processes: Near-infrared fluorescent microscopy, Single-molecular emission spectroscopy, Confocal microscopy, Mask aligner, Scanning electron Microscopy (SEM), transmission electron Microscopy (TEM), Polarized Optical Microscopy (POM), Clean room user

EXPERIENCE:

At Rice University, performed advanced research in understanding the attributes and behavior of worm-like flexible nanorods (single-walled carbon nanotube, SWCNT). The result can lead to the creation of "Reservoir Robot", which has high mobility and strong signal reflectance and can be used to find crude oil currently too-tiny-to-detect.

- Developed automated Matlab image processing code to track multiple worm-like flexible nanorods tracer for the first time to visualize flow in 10 times smaller published rock pore system with 10 times stronger signal intensity.
- Optimized Matlab code and leveraged parallel computing in cluster, achieving a reduction of video processing time from 15 hours to 30 minutes.
- Fabricated microfluidic based porous media platform from photolithography with 10 times smaller pore size of published setting to simulate currently undetected pores by packing silica colloids in channel.
- Obtained super-resolution nanotube curved chain backbone with 10x higher accuracy than current method by Gaussian fitting on image intensity in self-written Matlab code.
- Compared experiment results with 3-D Brownian dynamics simulation of bead-rod chain under confinement.
- Created bandpass-filtered near-infra red (NIR) microscopy to visualize SWCNT with specific flexibility to study the particle flexibility effect on mobility. This is the first time in the industry.

At Purdue University, research on core-shell magnetic nanoparticle delivery to cancer cell tissues

- Developed a brand new simple micro-scale lithography method for biological use.
- Applied self-synthesized core shell magnetic nanoparticle with special coating as anti-cancer drug delivery platform and NMR signal enhancer.

PUBLICATIONS ON PEER-REVIEWED JOURNALS:

- **Tang, Z.**; Wei, Q.; Wei, A. "*Metal-Mesh Lithography*". ACS Appl. Mater. Interfaces 2011, 3, 4812–4818.
- **Tang, Z.**; Wei, A. *"Fabrication of Anisotropic Metal Nanostructures Using Innovations in Template-Assisted Lithography"*. ACS Nano 2012, 6, 998–1003.
- Adnan, M.; Pinnick, R. A.; **Tang, Z.**; Taylor, L. W.; Pamulapati, S. S.; Carfagni, G. R.; Pasquali, M. "*Bending Behavior of CNT Fibers and Their Scaling Laws*". Soft Matter 2018, 14 (41), 8284–8292.
- Smith, A. D.; **Tang, Z.**; Pasquali, M.; Martí, A. *"Real-Time Visualization and Dynamics of Boron Nitride Nanotubes Undergoing Brownian Motion"*. J. Phys. Chem 2020, (accepted).
- **Tang, Z.**; Eichmann, S.L.; Lounis, B.; Cognet, L.; MacKintosh, F. C.; Pasquali, M. "*Semiflexible filament Brownian motion and molecular bending in complex media*". (In preparation).

PRESENTATIONS AT NATIONAL\INTERNATIONAL CONFERENCES:

OCT 2018 **Z. Tang,** S.L. Eichmann, R. Headrick, F. Mackintosh, M. Pasquali, "*Observation of semiflexible filament thermal bending and transport in complex porous media*", Society of Rheology annual meeting, 2018, Houston, TX.

NOV 2017 **Z. Tang,** S.L. Eichmann, R. Headrick, F. Mackintosh, M. Pasquali, "*Brownian Diffusion of Single Walled Carbon Nanotubes in Rock-like Colloidal Crystal Pores*", Nanoscale science and Engineering Forum session. AIChE, 2017, Minneapolis, MN.

AUG 2017 **Z. Tang,** S.L. Eichmann, R. Headrick, F. Mackintosh, M. Pasquali, "*SWNTs dynamics in rock-like porous media*", Texas Soft Matter Meeting, 2017, Houston, TX.

MAR 2017 **Z. Tang,** S.L. Eichmann, R. Headrick, F. Mackintosh, M. Pasquali, "*Brownian diffusion of single walled carbon nanotubes in highly confined rock-like porous media"*, APS Meeting, New Orleans, LA.

POSTERS AT NATIONAL/INTERNATIONAL CONFERENCES:

APR 2016 **Z. Tang,** S.L. Eichmann, M. Pasquali "*Individual SWNT as 'Reservoir-robot' for fluid control and sensing*", Chevron Lecture, Houston, TX

AWARDS AND ACHIEVEMENTS:

- Harrison M. Stine Memorial Scholarship, Purdue University, Summer 2010
- Undergraduate research fellowship from the Purdue Center for Cancer Research, Summer 2011
- Hin Wei Wong Graduate Fellowship Award from Rice University to recognize incoming students for outstanding achievements, Fall 2012.
- Saudi Aramco Research Fellowship Scholar, 2012-2015
- Robert Welch Research Fellowship, Scholar, 2016-2018

TEACHING EXPERIENCE:

2010-2013 Teaching Assistant, Rice University (4 semesters)

- Teaching Assistant: Numerical methods for differential equation in engineering and biology, Spring 2015.
- Teaching Assistant: Transport Phenomena I, Rice University, Fall 2012 and 2013.
- Teaching Assistant: Chemical Engineering Lab II, Rice University, Spring 2013.

ACTIVITIES:

2015-present Member of APS (American Physical Society)
2018-present Member of SoR (Society of Rheology)